FINITE TYPE SPACES OVER QCQS BASES ARE CLOSED IN FINITE PRESENTATION ONES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3049-3088, following the statement at 3035-3047.

Editorial note. No omissions; the monomorphism detection step reuses lemma-finite-type-eventually-closed (card L26290).

Allowed standard prerequisites (editorial).

- Approximation by finite type spaces. proposition-approximate (same chapter, published as problem 1743 tag 07SU, cited at ~3050-3052). X = lim X_i with affine transitions.
- Finite type over Z is finite presentation; base change. Morphisms of Spaces, Lemmas spaces-morphisms-lemma-noetherian-finite-type-finite-presentation and spaces-morphisms-lemma-base-change-finite-presentation (cross-chapter, cited at ~3060-3065). X_{i,Y} -> Y is of finite presentation.
- Monomorphisms appear at a finite stage. lemma-finite-type-eventually-closed (same chapter, card L26290, cited at ~3067-3068). X -> X_{i,Y} is a monomorphism for i large.
- Monomorphism of finite type is separated locally quasi-finite. Morphisms of Spaces, Lemmas spaces-morphisms-lemma-permanence-finite-type and spaces-morphisms-lemma-monomorphism-loc-finite-type-loc-quasi-finite (cross-chapter, cited at ~3069-3074). X -> X_{i,Y} separated and locally quasi-finite, hence representable.
- Quasi-affineness transfer principle. Spaces, Lemma spaces-lemma-representable-transformations-property-implication and More on Morphisms, Lemma more-morphisms-lemma-quasi-finite-separated-quasi-affine (cross-chapter, cited at ~3076-3080). X -> X_{i,Y} is quasi-affine.
- Quasi-affine closed in finite presentation. lemma-quasi-affine-closed-in-finite-presentation (same chapter, cited at ~3081-3084). Factorization X -> X' -> X_{i,Y} with X -> X' a closed immersion.
- Composition of finite presentation morphisms. Morphisms of Spaces, Lemma spaces-morphisms-lemma-composition-finite-presentation (cross-chapter, cited at ~3084-3087). X' -> Y is of finite presentation.

Source reference. Original lemma 0828: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1452–1452. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0871: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 3008–3008. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02LR: focused-seeds/geometry/S9-spaces-limits-finite-type-closed-in-finite-presentation/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 07SU: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2328–2328. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02YO: focused-seeds/geometry/S9-spaces-limits-finite-type-closed-in-finite-presentation/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03XR: focused-seeds/geometry/S9-spaces-limits-finite-type-closed-in-finite-presentation/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03XQ: focused-seeds/geometry/S9-spaces-limits-finite-type-closed-in-finite-presentation/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0463: focused-seeds/geometry/S9-spaces-limits-finite-type-closed-in-finite-presentation/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06G4: focused-seeds/geometry/S9-spaces-limits-finite-type-closed-in-finite-presentation/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0462: focused-seeds/geometry/S9-spaces-limits-finite-type-closed-in-finite-presentation/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume:*

(1) *f is of locally of finite type.*
(2) *X is quasi-compact and quasi-separated, and*
(3) *Y is quasi-compact and quasi-separated.*

Then there exists a morphism of finite presentation $f' : X' \to Y$ and a closed immersion $X \to X'$ of algebraic spaces over Y.

Proof. By Proposition 07SU we can write $X = \lim_i X_i$ with X_i quasi-separated of finite type over $\mathbf{Z}$ and with transition morphisms $f_{ii'} : X_i \to X_{i'}$ affine. Consider the commutative diagram

$$\begin{array}{ccccc} X & \longrightarrow & X_{i,Y} & \longrightarrow & X_i \\ & \searrow & \downarrow & & \downarrow \\ & & Y & \longrightarrow & \mathrm{Spec}(\mathbf{Z}) \end{array}$$

Note that X_i is of finite presentation over $\mathrm{Spec}(\mathbf{Z})$, see Morphisms of Spaces, Lemma 06G4. Hence the base change $X_{i,Y} \to Y$ is of finite presentation by Morphisms of Spaces, Lemma 03XR. Observe that $\lim X_{i,Y} = X \times Y$ and that $X \to X \times Y$ is a monomorphism. By Lemma 0828 we see that $X \to X_{i,Y}$ is a monomorphism for i large enough. Fix such an i. Note that $X \to X_{i,Y}$ is locally of finite type (Morphisms of Spaces, Lemma 0462) and a monomorphism, hence separated and locally quasi-finite (Morphisms of Spaces, Lemma 0463). Hence $X \to X_{i,Y}$ is representable. Hence $X \to X_{i,Y}$ is quasi-affine because we can use the principle Spaces, Lemma 02YO and the result for morphisms of schemes More on Morphisms, Lemma 02LR. Thus Lemma 0871 gives a factorization $X \to X' \to X_{i,Y}$ with $X \to X'$ a closed immersion and $X' \to X_{i,Y}$ of finite presentation. Finally, $X' \to Y$ is of finite presentation as a composition of morphisms of finite presentation (Morphisms of Spaces, Lemma 03XQ). □